



Original research

Effects of graduated compression socks on ankle inversion proprioception of half-marathon runners at different running distances

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ABSTRACT

Objectives: To determine the effect of graduated compression socks on ankle proprioception and the relationship between fatigue and ankle proprioception measures in half-marathon runners at successive 7 km intervals during a 21-km run.

Design: Repeated measures study.

Methods: Twenty well-trained half-marathon runners (mean $\pm$ SD: age 38.6 ± 11.3 y, weekly volume 42.3 ± 21.1 km, training experience 4.7 ± 3.5 y) performed two 21 km running trials on a treadmill wearing either graduated compression socks or looser, standard non-compression running socks. Each running trial was divided into three self-paced 7 km blocks, repeated at the same pace for each trial, heart rate was not controlled, but after each block, heart rate, blood lactate, rating of perceived exertion, and ankle inversion proprioception were measured, the latter by using the active movement extent discrimination apparatus.

Results: There was no differences between the graduated compression sock and non-compression running sock conditions in rating of perceived exertion scores ($F = 0.524, p = 0.478$), heart rate ($F = 0.001, p = 0.975$) and blood lactate ($F = 0.004, p = 0.951$). Overall, wearing graduated compression socks was associated with significantly better ankle proprioception after 21 km of running ($p = 0.024$, 95% CI = $-0.066, -0.005$). Heart rate, rating of perceived exertion and blood lactate were not correlated with ankle proprioception sensitivity in the graduated compression sock and non-compression running sock conditions at any point during the run (all $p > 0.05$).

Conclusions: Wearing graduated compression socks provides a positive effect on proprioceptive control of the ankle joint only after 21 km and may therefore reduce the possibility of injury associated with proprioceptive ability diminishing over time during exercise.

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Practical implications

- Compression socks were associated with better ankle proprioceptive sensitivity at 21 km, but not at 7 km and 14 km.
- There was no correlation between level of fatigue and ankle proprioception during half-marathon.
- Wearing GCSs can improve ankle proprioception, which may reduce ankle injury rates in half marathon runners.

1. Introduction

The half-marathon has become increasingly popular and it has become the most prevalent running event worldwide.¹ However, this popularity is closely followed by an increased rate of injury during the event.² It is reported that the highest incidence of lower limb injury (18.2%–92.4%) occurs in long-distance (4 km to a full marathon) runners.³ During running, the ground reaction force of the ankle is 3–4 times body weight.⁴ According to the relevant research, the ankle injury rate in the half-marathon is up to 6%,⁵ with ankle injury potentially being debilitating to an athlete's performance and detrimental to training adaptations.^{6,7} Therefore, intervention for the prevention of ankle injury during half-marathon running is particularly important.

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An increased risk of lower limb injury is associated with decreased ankle proprioception.⁸ Proprioception is the ability to control the movement and position of a limb in space by integrating the sensory information gathered by mechanical receptors.⁹ In many sports, such as running, basketball, football, and soccer, the ankle-foot complex is the only part of the body in contact with the ground in the process of sport participation.⁸ Therefore, ankle proprioception provides the runner with important information about adjusting the ankle position and coordinating body movements.⁸ Secondly, ankle proprioception is considered to be one of the intrinsic risk factors for ankle injury, with its decline often leading to further ankle injury.¹⁰ Therefore, finding a way to minimize potential reduction in ankle proprioception may be useful for injury prevention. The main receptors that produce proprioceptive information are located in muscles, tendons, ligaments, and joint capsules, while receptors in the deeper layers of the skin and the muscle fascia are considered to be complementary sources.¹¹ Accordingly, using an intervention that increases sensory input by stimulating skin receptors may be effective in enhancing proprioception and reducing the incidence of ankle injury.

In recent years, there has been an increase in the use of GCSs as an aid to training.^{6,12} There is evidence to suggest that compression materials can be used to enhance tactile stimulation to the receptors in the skin surface and joint soft tissues, thereby introducing additional intermittent sensory input in terms of proprioceptive information, and enhancing the perception of somatosensory feedback.¹³ Graduated compression socks (GCSs) used as an auxiliary training tool could compress the skin surface of the marathoners' lower limbs, which may improve proprioceptive information input.¹³ In addition, compression garments have been reported to effectively reduce soft tissue vibration and muscle activity associated with the impact experienced during running, thereby reducing fatigue¹² and, potentially, attenuating decrements in proprioception.

It is expected that fatigue within the last tertile of races would have similar associations with increased injury risk. The biomechanics and neuromuscular control ability of participants in endurance races will deteriorate as the running distance increases¹⁴ and these changes have been linked to injury risk.¹⁵ In addition, it is speculated that increased injury rate may be linked to changes in proprioception caused by accumulated fatigue.¹⁶ Compression garments have been reported to effectively reduce soft tissue vibration and muscle activity associated with the impact experienced during running, thereby reducing fatigue¹² and, potentially, attenuating decrements in proprioception.

However, no previous research has investigated the effect of GCSs on ankle proprioception during prolonged running exercise, with different levels of fatigue assessed by blood lactate and ratings of perceived exertion. Therefore, the aims of this study were to i) determine the relationship between the level of fatigue and ankle proprioception and ii) investigate the effect of GCSs on ankle proprioception after each of three equal segments of a half-marathon distance run. We hypothesized that: (1) level of fatigue is associated with ankle proprioception decline during running, and (2) compared to normal socks, GCSs can work to retain ankle proprioception during prolonged running.

2. Methods

The study was approved and followed the ethical guidelines of the research ethics committee of Shanghai University of Sport (approval number 102772020RT069). All participants were volunteers and signed informed consent before data collection. The G*power software was used to calculate the required sample size. With an anticipated effect size = 0.3, significance level at $p = 0.05$, and a power = 0.8, 17 participants were required for the study. Considering the attrition rate to be 15%, 20 participants were eventually recruited through public social media. The inclusion criteria for the participants were: (1) aged between 18 and 45; (2) each participant engaged in regular training and having participated in a half-marathon in the past 6 months; (3) no

lower limb injuries sustained in the past six months; (4) had never used GCSs before; and (5) agreed to do the tests for the experiment and signed the informed consent. The exclusion criteria were: (1) lower limb injuries occurred in the past six months; (2) previous lower limb surgeries; (3) serious skin diseases and unhealed scars; and (4) chronic ankle instability (Cumberland Ankle Instability Tool (CAIT) score ≤ 24 ¹⁷). The details of the participants' characteristics are presented in Table 1.

The non-compression loose socks (NCSs) were each runner's preferred running socks. Graduated compression socks, GCSs by CEP (Germany) were used in the other running condition. The GCS materials are 69% polyamide, 16% spandex, and 15% polypropylene. According to the parameters provided by CEP, the average compression decreased from 22 mm Hg at the ankle to 18 mm Hg successively. The appropriate GCS size was determined by measuring the circumference at the narrowest part of each participant's ankle using the manufacturer's sizing chart.

During running, participants' heart rate was continuously monitored (Polar H10, Finland) and reported rating of perceived exertion (RPE) collected. Perceived fatigue was measured by using the Borg RPE scale which can also represent the level of fatigue,¹⁸ ranging from 6 (no exertion at all) to 20 (maximal exertion). Blood lactate concentration (mmol/L) was measured by the finger prick method (Lactate Scout4, Germany), as an objective marker of fatigue.¹⁹

The active movement extent discrimination apparatus (AMEDA) was used to assess ankle inversion proprioception, as it has shown to be sensitive to a variety of injuries that affect active ankle proprioception.²⁰ During measurement, only the dominant foot (Waterloo Footedness Questionnaire) was tested. Previous research indicates that the dominant side has poorer proprioceptive ability.²¹ Therefore, given the ultimate goal of this study (prevent ankle injury), it is necessary to test the ankle joint on the side with worse proprioception as representing the participant's proprioception ability. The AMEDA consists of a stationary platform and a moveable test platform. The moveable test platform is equipped with a board that rotates along a central axis to produce a tilted foot angle. The inversion movement of the board can be stopped at set angles to produce 4 labeled positions 1, 2, 3, and 4, with position 1 = 10°, position 2 = 12°, position 3 = 14°, and position 4 = 16°. Participants were asked to stand upright with their eyes straight forward and their feet on a platform with the middle toe of the test foot aligned with the central axis of the moveable test platform so that the test foot could freely perform ankle inversion movement at normal speed. During the test, they performed inversion movements of the testing foot according to the instructions, judged the position after reaching the inversion testing position (b), and returned to the starting position (a); the other foot was placed on a fixed platform and did not move during the test; at the beginning of the test, the participants were first familiarized with the 4 positions, by feeling each position 1, 2, 3, and 4 in order, 3 times through. During the test, the 4 different inversion positions were presented in random order 10 times each, and participants were required to make an absolute judgment regarding the number of the position they felt each time (Fig. 1).

The experiment consisted of two trials with each participant running 21 k with a one-week interval between. The testing sequence was

Table 1
Participant characteristics (mean $\pm$ SD).

Variable	Men (N = 10)	Women (N = 10)	Total (N = 20)
Age (y)	34.1 $\pm$ 10.7	43 $\pm$ 10.5	38.6 $\pm$ 11.3
Height (cm)	175.9 $\pm$ 6.8	162.7 $\pm$ 9.2	169.3 $\pm$ 10.4
Body mass (kg)	67.3 $\pm$ 7.7	53.4 $\pm$ 4.6	60.3 $\pm$ 9.5
BMI (kg·m ⁻²)	21.7 $\pm$ 1.8	20.8 $\pm$ 1.7	21.3 $\pm$ 1.8
Weekly volume (km)	46.5 $\pm$ 22.5	38 $\pm$ 19.9	42.3 $\pm$ 21.1
Training experience (y)	4.4 $\pm$ 2.1	5 $\pm$ 4.6	4.7 $\pm$ 3.5

SD = standard deviation.

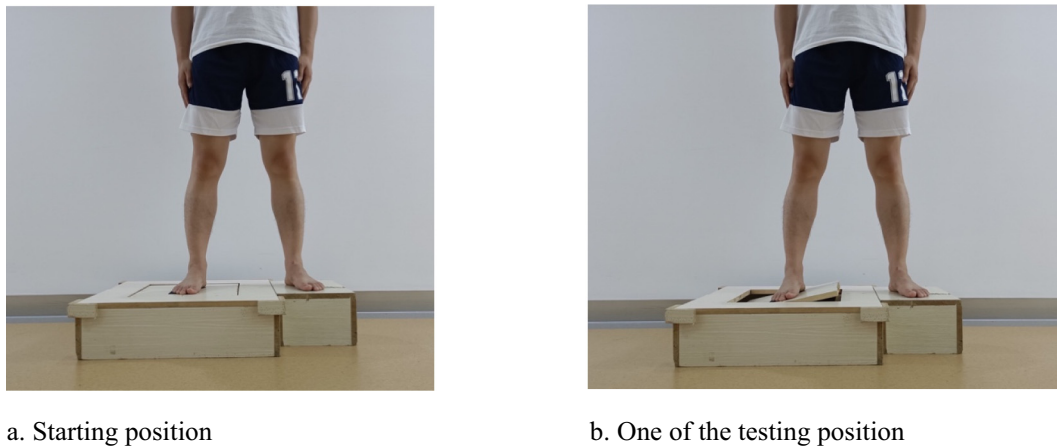


Fig. 1. Ankle inversion proprioception assessed by the AMEDA.

Table 2

Completion time of each 7 km in GCSs and NCSs (mean $\pm$ SD).

Distance	GCSs (min)	NCSs (min)	<i>p</i>	95% CI
The 1st 7 km	39.3 $\pm$ 4.1	39.5 $\pm$ 4.3	0.530	−0.855, 0.455
The 2nd 7 km	39.6 $\pm$ 6.1	39.7 $\pm$ 6.2	0.867	−0.666, 0.566
The 3rd 21 km	40.2 $\pm$ 4.3	39.5 $\pm$ 3.9	0.073	−0.067, 1.367
Total	119.2 $\pm$ 13.3	118.6 $\pm$ 13.2	0.319	−0.628, 1.023

randomized using the random code in Python 3.7. To consider the effect of temperature, both running tests were conducted indoors at a temperature of 14 °C on a treadmill. During the experiment, participants were instructed to run at a self-paced speed²² as per their average training experience in the past month, with similar completion time (Table 2). Each running test was divided into three segments, each of which was 7 km in length, and the total distance was 21 km. During the run, we monitored the heart rate of each participant, collected the RPE, blood lactate and tested the ankle inversion proprioception before running and after each 7 km block, each of which took approximately 5 min (Fig. 2).

IBM SPSS Statistics (<https://www.ibm.com/products/spss-statistics> IBM Corporation, USA. Version 25) was used for data analysis, and $p < 0.05$ indicated that the results were statistically significant. Ankle inversion proprioceptive performance was represented by the area under the receiver operating curve (AUC), for a participant's ability to discriminate between adjacent positions on the test, and the mean of the 3 AUC

scores was calculated for their total score. AUC values can range from 0.5 (chance responding) to 1.0 (perfect discrimination). Two-way repeated measures ANOVA was used to determine the effects of GCSs on ankle inversion proprioception with different running distances, and the difference of heart rate and RPE scores between conditions. The relationship between RPE, heart rate, blood lactate and ankle inversion proprioception was analyzed using the Pearson correlation analysis. Paired sample *t*-tests were used for post-hoc analysis to examine the differences between conditions.

3. Results

As the distance increased, both RPE and heart rate increased curvilinearly, with an initial sharp increase followed by a plateau (Fig. 3a & b), indicating progressive fatigue effects. Repeated measures ANOVA showed that there was no difference in RPE ($F = 0.524$, $p = 0.478$), heart rate ($F = 0.001$, $p = 0.975$), and blood lactate ($F = 0.004$, $p = 0.951$) between GCS and NCS conditions, and a similar pace was observed in both conditions reflecting a similar trend in degree of fatigue for both conditions (Fig. 3a & b). Pearson correlation analysis showed that heart rate, RPE and blood lactate were not correlated with ankle proprioception in the GCS and NCS conditions before running and after each 7 km (all $p > 0.05$).

Repeated measures ANOVA showed that there was no interaction effect between time and condition ($F = 0.962$, $p = 0.417$), and there was not a significant main effect for time ($F = 0.151$, $p = 0.702$) and

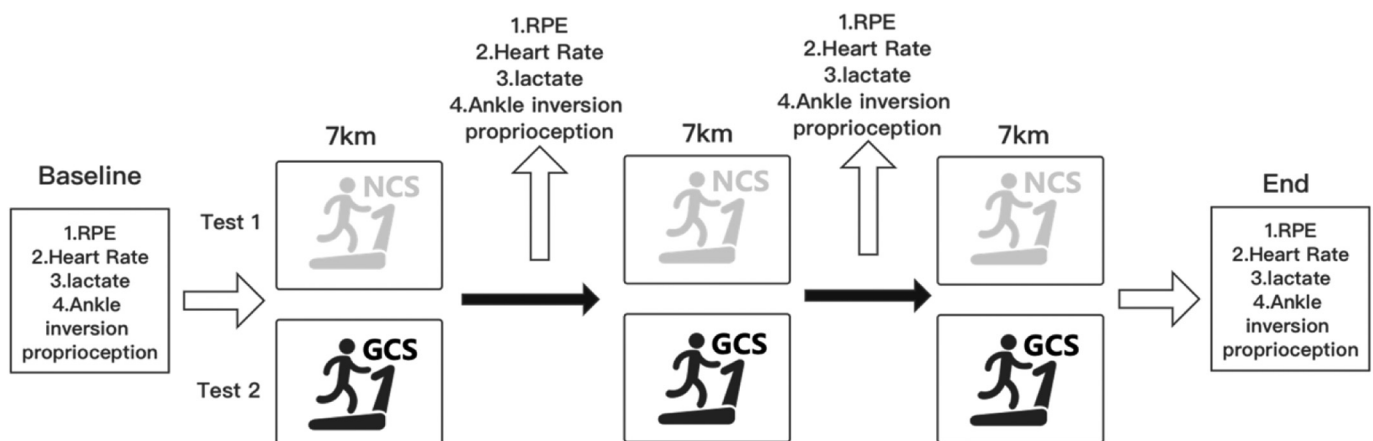
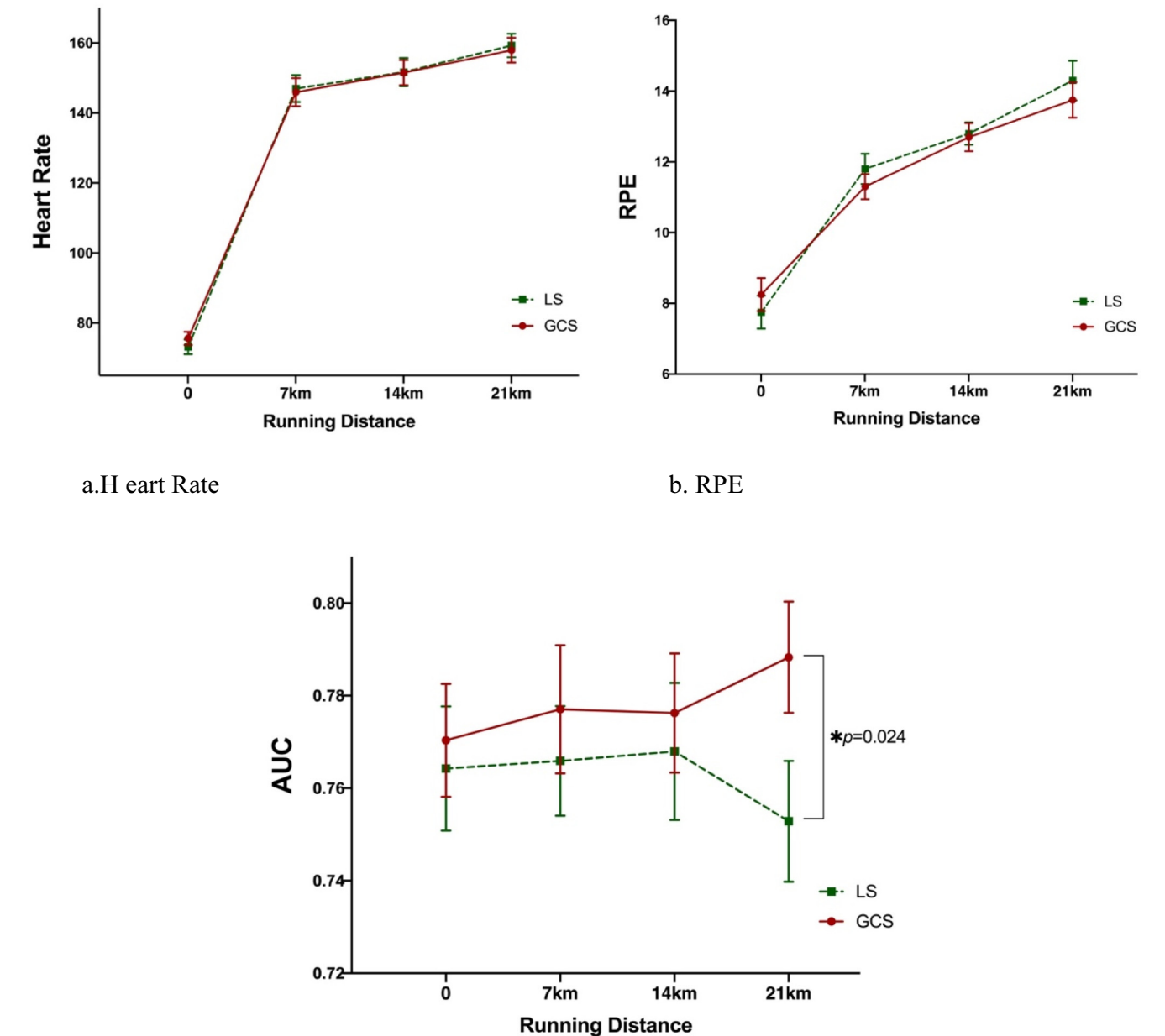


Fig. 2. Experimental flow diagram.



c. Ankle inversion proprioception. * = $p < 0.05$; AUC=area under the receiver operating characteristic curve

Fig. 3. Heart rate, RPE and ankle inversion proprioception with different running distances in the GCS and NCS conditions. GCS = graduated compression sock, NCS = non-compression sock.

condition ($F = 2.198$, $p = 0.155$). According to the results of paired t -test, there was no significant difference in total running time ($p = 0.151$, 95% CI = -1.428 , 0.228 , Cohen's $d = 0.05$) between the GCS condition and the NCS condition, and also no statistically significant

difference in ankle proprioception between the two conditions before running ($p = 0.741$, 95% CI = -0.044 , -0.032 , Cohen's $d = 0.1$), after 7 km ($p = 0.528$, 95% CI = -0.047 , 0.025 , Cohen's $d = 0.19$) or after 14 km ($p = 0.519$, 95% CI = -0.035 , 0.018 , Cohen's $d = 0.13$), but after

Table 3
Differences of ankle AUC between GCS condition and NCS condition at different running distances.

Distance	GCS AUC (mean $\pm$ SD)	NCS AUC (mean $\pm$ SD)	Mean difference (mean $\pm$ SEM)	<i>p</i>	95% CI	Cohen's <i>d</i>
0	0.770 $\pm$ 0.055	0.764 $\pm$ 0.060	−0.006 $\pm$ 0.018	0.741	−0.044, −0.032	0.1
7 km	0.777 $\pm$ 0.062	0.766 $\pm$ 0.053	−0.011 $\pm$ 0.017	0.528	−0.047, 0.025	0.19
14 km	0.776 $\pm$ 0.058	0.768 $\pm$ 0.066	−0.008 $\pm$ 0.013	0.519	−0.035, 0.018	0.13
21 km	0.788 $\pm$ 0.054	0.753 $\pm$ 0.058	−0.035 $\pm$ 0.014	0.024*	−0.066, −0.005	0.62

GCS = graduated compression sock, NCS = non-compression sock, SD = standard deviation, SEM = standard error of mean.
* $p < 0.05$.

21 km, at the last measurement, ankle proprioception was significantly better when compression socks were worn ($p = 0.024$, 95% CI = -0.066 , -0.005 , Cohen's $d = 0.62$) (Table 3 & Fig. 3c).

4. Discussion

Data obtained at the successive 7 km running blocks show that RPE, heart rate and blood lactate all had an upward trend in the two 21 km running tests, but at no point were significantly correlated with ankle proprioceptive sensitivity. While no significant difference was found in AUC sensitivity scores between the two conditions at 7 km and 14 km, after running 21 km, ankle proprioception was significantly better in the GCS condition than in the NCS condition. Data obtained at the successive 7 km running blocks show that RPE, heart rate and blood lactate all had an upward trend in the two 21 km running tests, but at no point were significantly correlated with ankle proprioceptive sensitivity, which was in contradiction of our first hypothesis. While no significant difference was found in AUC sensitivity scores between the two conditions at 7 km and 14 km, after running 21 km, ankle proprioception was significantly better in the GCS condition than in the NCS condition. Therefore, our second hypothesis was supported.

In this study, although indications for fatigue were becoming more apparent, this did not affect ankle proprioception in well-trained runners. This finding is different from previous studies of knee and shoulder joints and proprioception.^{23,24} One possible explanation for this is that the spindles of the muscles crossing the ankle do not fatigue with their motor counterparts. If so, the afferent nervous system may remain viable even in the case of severe motor neuron fatigue.²⁵ It is also possible that the ankle joint relies more on proprioceptive components within the skin, ligaments and joint capsules²⁵ than on the muscle spindles that play a major role in proprioception.¹¹ In addition to the above possibilities, the regular training of the participants may have been a factor. A previous study showed that regular physical activity can lead to enhanced proprioception by modulating muscle spindles and inducing plasticity changes in the central nervous system.¹⁶ Therefore, proprioception improved in this way may be less affected by fatigue.

Wearing GCSs was associated with better ankle proprioception when the running distance reaches 21 km, but no significant effect was observed after 7 km and 14 km. This finding may be related to the GCSs effectively reducing the muscle vibration caused by foot–ankle contact with the ground.⁶ Manzone et al.²⁶ confirmed that muscle vibration could affect proprioception. Meanwhile, Hausswirth et al.¹⁸ found that muscle amplitude increased significantly after running for 1 h and 50 min. Given the average completion time of runners in this study, they were finishing running the third 7 km after 1 h and 50 min. Therefore, the effect of GCSs on proprioception after the third 7 km run may be related to the decrease in muscle amplitude 1 h and 50 min later. The GCS condition permitted this ankle proprioceptive sensitivity maintenance despite fatigue, while the NCS condition could not.

In contrast to the current findings, another study found that compression garments did not improve joint position sense when the threshold to detection of passive motion (TTDPM) test was used to measure proprioception.²⁷ One reason for these opposite results may be that proprioception testing methods are different, as they have different ecological validities and sensitivities. Moreover, Ashton-Miller et al.²⁸ found that the slow movement speed would make some proprioceptive receptors unable to be stimulated, potentially contributing to poorer judgment of joint position. To optimize the performance of activities of daily life and sport, the limbs need to be active and functioning within normal movement speeds. Magill et al.²⁹ using a TTDPM (the threshold to detection of passive motion) proprioception assessment found that the movement termination position is important for the accuracy of the limb's position judgment because most exercise events are terminated by external factors, such as the ankle angle when the foot is touching the ground during running and so on. Therefore, TTDPM assessment may not be sensitive enough to show effects on proprioception, or it may produce results

that differ depending on whether the participant was weight-bearing during the test. Incorporating the sense of ankle position into the test at the termination of a movement can improve ecological validity, and the AMEDA incorporates movement termination, allows normal weight-bearing and active movement of joints, and the speed of joint movement is close to activities of daily life. By potentially optimizing the sense of joint position at movement termination, this may enhance the central nervous system collection and integration of proprioceptive information under noise interference, meaning that the test has higher ecological validity and sensitivity.³⁰

An important implication of this study is that the GCSs have different effects on ankle proprioception than conventional running socks in well-trained runners over a half-marathon distance. Future research could extend the running distance, for example to marathon distance, to observe whether GCSs have the ability to maintain proprioception for longer running load.

There are some limitations to the study. First, in order to control the impact of environmental temperature in this study, participants were asked to complete the 21 km running test on a treadmill, in an indoor environment. It may also be worthwhile assessing runners at distances from 14 to 21 km to determine the threshold distance at which the GCSs begin to protect against reductions in proprioceptive performance. Future research should aim to assess running on the road in as controlled conditions as possible.

5. Conclusion

Wearing graduated compression socks was associated with better maintenance of ankle inversion proprioception beyond the 14 km distance mark of a half-marathon run when compared to non-compression socks. Heart rate, RPE, and blood lactate were not significantly correlated with ankle proprioceptive sensitivity, suggesting that fatigue did not affect ankle proprioception in half-marathon runners. Based on these findings, wearing GCSs in half-marathon running may reduce the risk of ankle injuries.

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Declaration of interest statement

The work described here is original research that has not been published previously, and is not under consideration for publication elsewhere, either in whole or in part. This manuscript has no conflict of interest and has been approved for publication by all authors.

Confirmation of ethical compliance

All procedures performed in this study involving human participants were in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

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